

Veltrixa Cloud

Data Flow Mapping

AI Document Classification Service

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Document Control

0.1 Document Information

Field	Value
Document Title	Data Flow Mapping – AI Document Classification
Document ID	VG-PGO-DFM-001
Status	Final
Business Owner	Product Management
Technical Owner	Cloud Engineering
Review Frequency	Annual or major architectural change

0.2 Version History

Version	Date	Author	Description
0.1	July 2026	Privacy Governance Analyst	Initial master document
1.0	July 2026	Privacy Governance Analyst	Portfolio Edition approved

0.3 Approvals

Role	Status
Lead Solution Architect	Approved
Senior Privacy Counsel	Approved
Head of Privacy	Approved

0.4 Related Documents

- Privacy Discovery Questionnaire
- Data Flow Diagram
- Privacy by Design Review
- DPIA Screening
- ROPA
- Vendor Reviews
- Transfer Impact Assessment

1. Introduction

1.1 Executive Summary

Veltrixa Cloud is a fictional enterprise SaaS platform that provides AI-assisted document management. This Data Flow Mapping documents how personal data enters, moves through, is stored within and ultimately is deleted from the AI Document Classification capability. It provides the factual foundation for Privacy by Design Reviews, DPIA screening, ROPA maintenance and vendor assessments.

1.2 Purpose

The purpose of this document is to establish a shared understanding of the complete lifecycle of personal data within the solution and to support engineering, security, privacy and product teams when assessing architectural changes.

1.3 Scope

In Scope

Document upload, validation, malware scanning, AI classification, metadata generation, storage, access management, logging, retention and deletion.

Out of Scope

Customer business processes outside Veltrixa Cloud and customer legal bases for processing.

Assumptions

Solution hosted in AWS Frankfurt; AI processing remains within the same environment.

Constraints

Customers may upload documents containing special category personal data; governance therefore focuses on secure processing rather than restricting uploads.

1.4 Intended Audience

- Privacy Governance
- Engineering
- Security
- Product Management
- Internal Audit
- Compliance

1.5 Applicable Regulations & Standards

Framework	Relevance
GDPR	Articles 5, 25, 30, 32 and 44–49
ISO/IEC 27001	Supports the ISMS by documenting information flows
ISO/IEC 27701	Supports the Privacy Information

	Management System
NIST Privacy Framework	Supports identification and communication of privacy risks

2. Business Context

2.1 Business Overview

Veltrixa Cloud is a fictional enterprise Software-as-a-Service platform designed to help organisations manage business documents and automate document-centric workflows. The platform is intended for medium and large organisations operating primarily within the European Union.

2.2 Business Process

Authorised users upload business documents through a secure web application. The platform validates each document, performs malware scanning, analyses the content using an internally hosted AI Classification Service and proposes document types and metadata. Users remain responsible for reviewing and approving all AI-generated suggestions before they become part of downstream business processes.

2.3 Data Subjects

Data Subject	Typical Examples
Employees	Employment contracts, HR records
Job applicants	CVs, cover letters
Customers	Business correspondence, agreements
Supplier contacts	Contracts, invoices
Consultants	Statements of work
Other individuals	References, emergency contacts, meeting participants
Children (where applicable)	Benefits, education or family-related documents

2.4 Categories of Personal Data

Category	Examples
Identity	Name, address, email, telephone number
Employment	Job title, employer, employee ID
Financial	Invoice details, payment references
Technical	User ID, IP address, timestamps
Special Category (possible)	Health data, trade union membership, political opinions, religion, ethnicity, biometric/genetic data

3. Solution Overview

3.1 System Overview

The solution consists of a secure web portal, an Upload Service, an encrypted temporary processing area, an internal AI Classification Service, Amazon S3 document storage, PostgreSQL metadata storage, a logging platform and role-based identity management.

3.2 Architecture Overview

The architecture separates document content from operational metadata and processes uploaded files entirely within the AWS Frankfurt region. The AI service performs analysis without using customer documents for model training and returns metadata suggestions to the application.

3.3 High-Level Architecture Diagram

3.3 High-Level Architecture Diagram

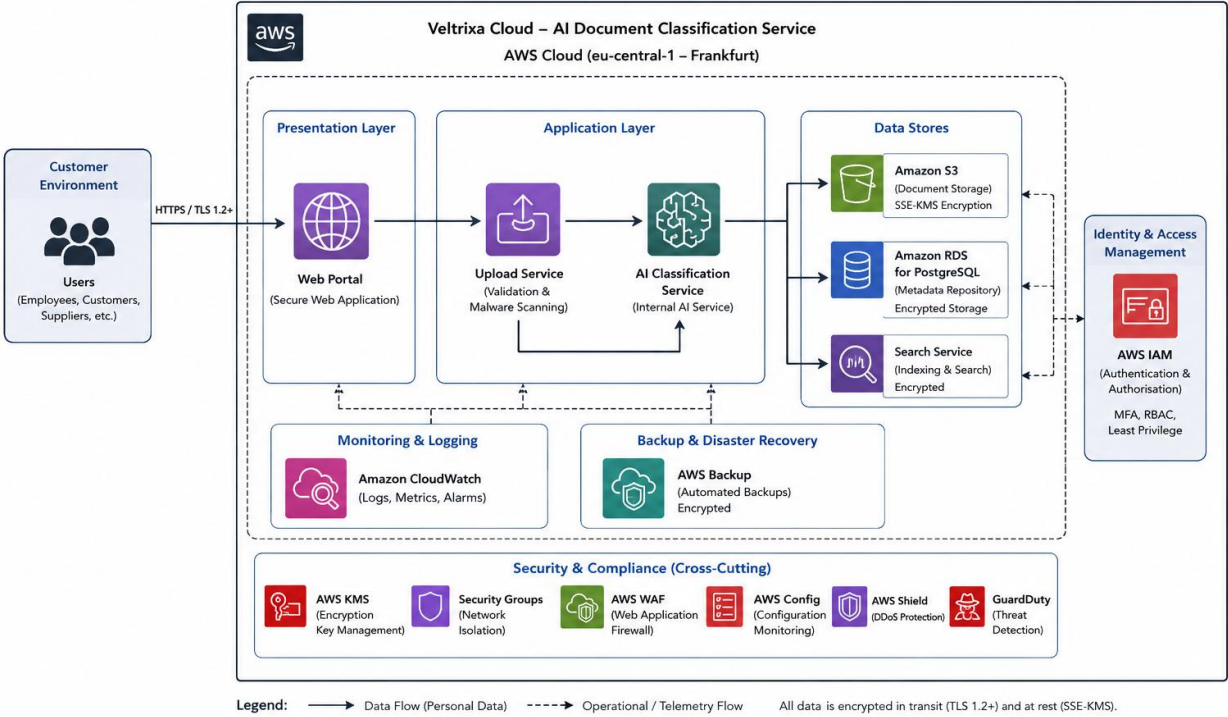


Figure 1. High-Level Architecture of the Veltrixa Cloud AI Document Classification Service.

Figure 1. High-Level Architecture of the Veltrixa Cloud AI Document Classification Service.

3.4 Core Components

Component	Purpose	Processes Personal Data
Web Portal	Document upload and user interaction	Yes
Upload Service	Validation and malware scanning	Yes
AI Classification Service	Classification and metadata generation	Yes
Amazon S3	Original document storage	Yes
PostgreSQL	Metadata repository	Yes

Logging Platform	Security monitoring	Limited technical data
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4. Data Flow Mapping

This chapter describes the complete lifecycle of personal data within the AI Document Classification capability. Each processing stage identifies the business purpose, the personal data involved, where the data resides, who can access it, the primary security controls in place and any notable privacy considerations. The objective is to provide a shared understanding of how personal data moves through the solution from initial upload to final deletion.

Figure 2 provides a high-level overview of the end-to-end personal data flow. The diagram illustrates the relationship between the principal processing stages, supporting services and data stores before each stage is examined in detail.

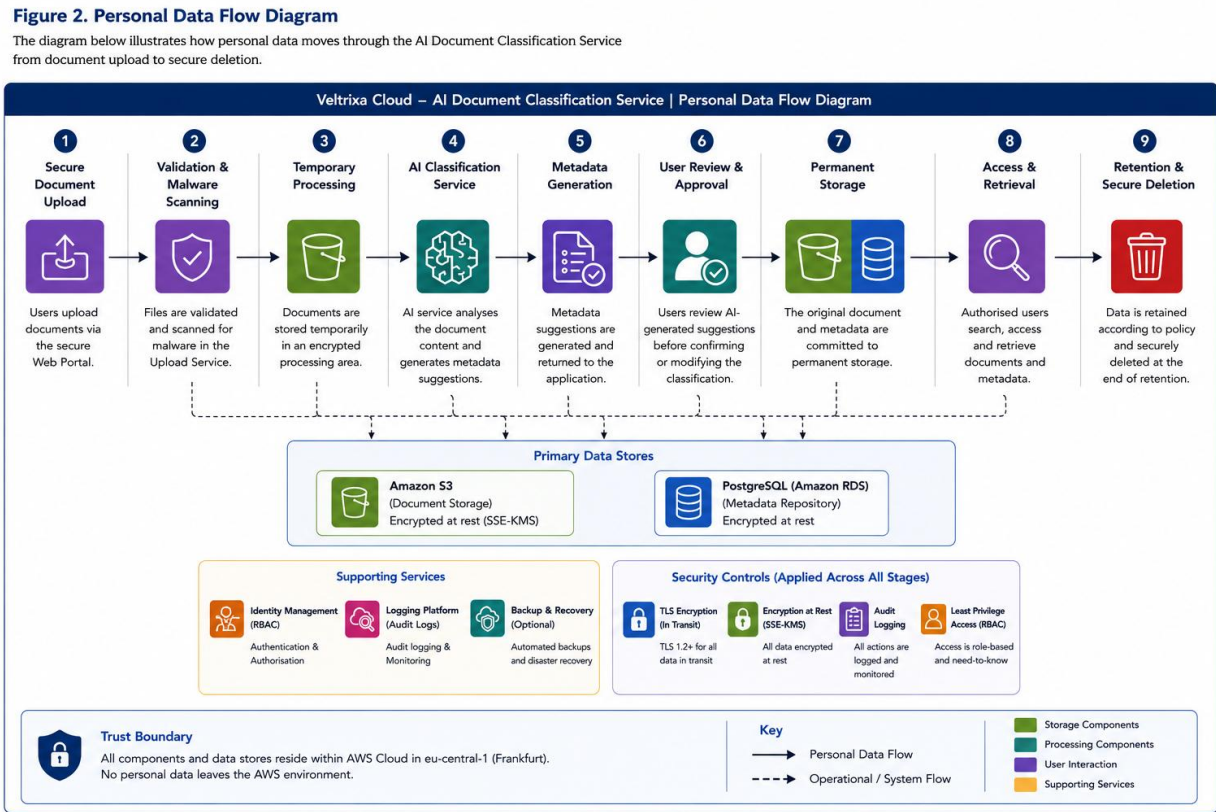


Figure 2. Personal Data Flow Diagram
End-to-end flow of personal data through the Veltrixa Cloud AI Document Classification Service, from document upload to secure deletion.

4.1 Stage 1 – Secure Document Upload

Business Purpose	Receive documents from authenticated customer users.
Description	Users upload supported documents through the secure web portal using an encrypted

	HTTPS/TLS connection.
Personal Data	Any personal data contained within the uploaded document together with technical upload metadata.
Data Location	Encrypted transmission channel.
Actors / Access	Authenticated customer user and Upload Service.
Security Controls	TLS encryption, authentication, file size validation.
Privacy Observation	The platform collects only the information required to receive and process the uploaded document.

4.2 Stage 2 – Validation & Malware Scanning

Business Purpose	Verify that uploaded files are safe and supported.
Description	The Upload Service validates the file type, file size and scans the document for malicious content before processing continues.
Personal Data	Original document and technical metadata.
Data Location	Temporary encrypted processing environment.
Actors / Access	Upload Service only.
Security Controls	Malware scanning, validation rules, restricted service permissions.
Privacy Observation	Validation reduces security risks before personal data enters downstream processing.

4.3 Stage 3 – Temporary Processing

Business Purpose	Prepare the document for AI analysis.
Description	The validated document is stored temporarily while processing takes place.
Personal Data	Original document.
Data Location	Encrypted processing bucket in AWS Frankfurt.
Actors / Access	Internal processing services.
Security Controls	Encryption at rest, lifecycle deletion, no backup.
Privacy Observation	Temporary copies are automatically removed after successful processing.

4.4 Stage 4 – AI Classification

Business Purpose	Identify document type and propose
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	metadata.
Description	The internal AI Classification Service analyses document structure and textual content to generate classification suggestions.
Personal Data	Document content.
Data Location	Internal AI processing service.
Actors / Access	Automated service only.
Security Controls	Internal deployment, no external AI providers, no customer model training.
Privacy Observation	The AI supports human decision-making and does not make binding decisions.

4.5 Stage 5 – Metadata Generation

Business Purpose	Generate searchable business metadata.
Description	The AI returns document type, tags and suggested metadata to the application.
Personal Data	Metadata associated with the uploaded document.
Data Location	Application services and PostgreSQL.
Actors / Access	Application and authorised users.
Security Controls	Role-based access control and audit logging.
Privacy Observation	Metadata is intentionally stored separately from the original document.

4.6 Stage 6 – User Review

Business Purpose	Maintain human oversight.
Description	Users review AI-generated suggestions before confirming or modifying the classification.
Personal Data	Metadata and document information.
Data Location	Business application.
Actors / Access	Authorised customer users.
Security Controls	Authentication, RBAC, audit trail.
Privacy Observation	Human review helps prevent incorrect automated classifications.

4.7 Stage 7 – Permanent Storage

Business Purpose	Store business records.
Description	The original document and metadata are committed to permanent storage.
Personal Data	Original documents and associated metadata.
Data Location	Amazon S3 and PostgreSQL (AWS Frankfurt).
Actors / Access	Authorised users according to RBAC.

Security Controls	Encryption at rest, RBAC, audit logging.
Privacy Observation	Separation of storage supports least-privilege access.

4.8 Stage 8 – Retention & Secure Deletion

Business Purpose	Ensure compliant retention of business records.
Description	Customer-configured retention rules determine when documents are securely deleted.
Personal Data	Documents, metadata and related logs.
Data Location	Storage services and logging platform.
Actors / Access	System services and authorised administrators.
Security Controls	Retention policies, lifecycle management, secure deletion.
Privacy Observation	Personal data is retained no longer than necessary for the configured business purpose.

5. Data Inventory

This inventory summarises the principal categories of personal data processed by the AI Document Classification capability together with the business purpose, storage location, primary access rights and retention approach.

Data Element	Category	Purpose	Source	Storage	Primary Access	Retention
Original document	Business record	Classification & storage	Customer upload	Amazon S3	Authorised users	Customer policy
Document metadata	Operational	Search & workflow	AI service	PostgreSQL	Application/RBA C	Customer policy
User account	Identity	Authentication	Identity provider	Identity platform	Administrators	Identity policy
Audit log	Technical	Security monitoring	Platform	Logging platform	Security team	Logging policy
Temporary processing copy	Transient	AI processing	Upload service	Encrypted processing bucket	Internal services	<1 minute

6. Data Stores

The solution deliberately separates business documents, operational metadata and security logs to support least-privilege access and reduce unnecessary exposure of personal data.

Repository	Contents	Region	Encryption	Backup	Owner
Amazon S3	Original documents	Germany (Frankfurt)	AES-256	Yes	Platform
PostgreSQL	Metadata	Germany (Frankfurt)	Encrypted	Yes	Platform
Processing Bucket	Temporary documents	Germany (Frankfurt)	Encrypted	No	Platform
Logging Platform	Audit logs	EU	Encrypted	Policy-based	Security

7. Data Transfers

7.1 Internal Transfers

Data moves internally between the Web Portal, Upload Service, temporary processing environment, AI Classification Service, Amazon S3, PostgreSQL and the business application over authenticated and encrypted channels.

7.2 External Transfers

Operational monitoring information may be transmitted to approved monitoring services. Customer document content is not intentionally shared with external AI providers.

7.3 Cross-Border Transfers

Normal processing remains within the EU. Any future international transfers require assessment under the organisation's vendor governance and transfer assessment process.

8. Trust Boundaries

The architecture crosses several logical trust boundaries that require appropriate technical and organisational controls.

Figure 3 provides an overview of the trust boundaries within the Veltrixa Cloud AI Document Classification Service. The diagram illustrates the principal security zones and the trust relationships between the customer environment, internal platform components, data stores and operational support services.

Figure 3. Trust Boundary Diagram

4.2 Trust Boundary Diagram

The diagram below identifies logical trust boundaries between the customer environment, the AWS cloud environment, internal services and third-party services.

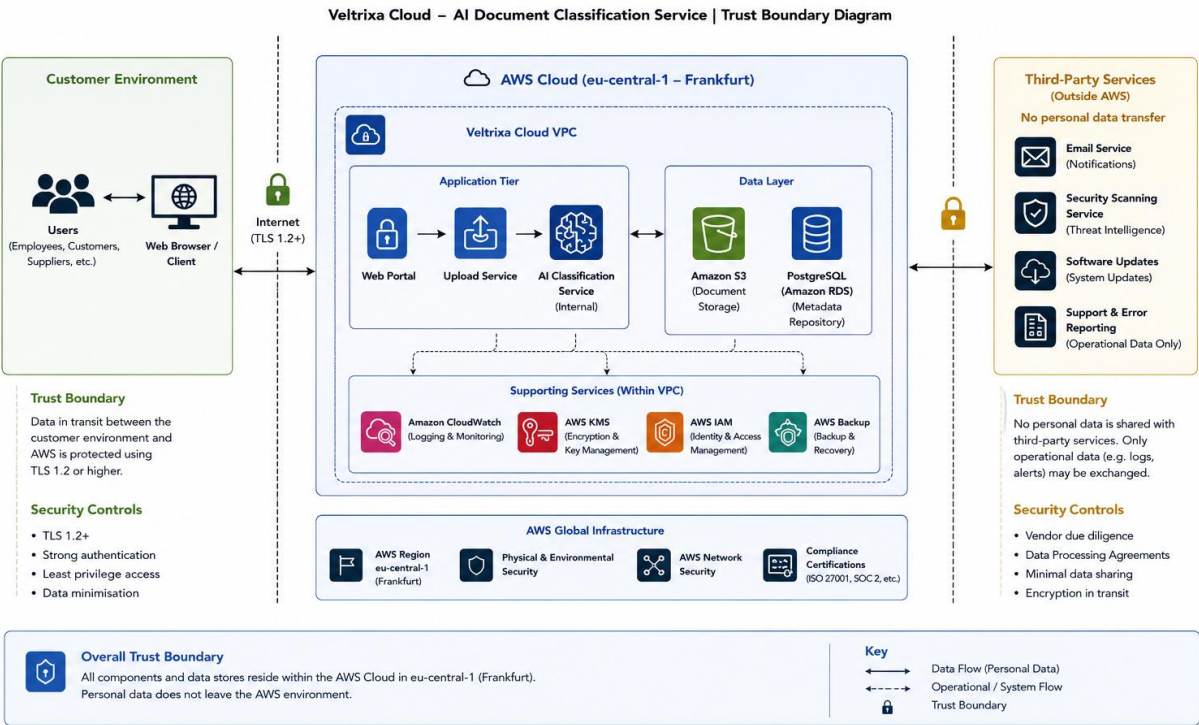


Figure 3. Trust Boundary Diagram. Logical trust boundaries within the Veltrixa Cloud AI Document Classification Service.

Boundary	Description	Primary Controls
Customer ↔ Platform	Public internet	TLS, authentication
Application ↔ AI	Internal processing	Service authentication
Application ↔ Storage	Persistent storage	RBAC, encryption
Platform ↔ Monitoring	Operational telemetry	Least privilege
Support ↔ Customer Tenant	Exceptional support	Approval and audit logging

9. Access Model

Access to personal data is governed using Role-Based Access Control (RBAC) and the principle of least privilege.

Role	Read	Write	Delete	Administrative Rights
Customer User	✓	Own content	No	No
Customer Administrator	✓	✓	Policy-based	User administration
Support Engineer	Temporary	No	No	Approved support only
Cloud Operations	Infrastructure only	No	No	Platform administration
Security Team	Logs only	No	No	Security monitoring

10. Third Parties

Veltrixa Cloud relies on a limited number of carefully selected third-party service providers. Each provider is subject to vendor due diligence, contractual controls and periodic privacy and security reviews.

Vendor	Service	Role	Personal Data	Region	Key Safeguards
AWS	Infrastructure Hosting	Processor	Documents, metadata	Germany (Frankfurt)	DPA, encryption, RBAC
Datadog	Monitoring & Logging	Sub-processor	Technical telemetry	EU / contractual controls	Vendor review, access controls
Identity Provider	Authentication	Processor	User identity	EU	Federated authentication, MFA

11. International Transfers

11.1 Assessment

The target architecture is designed so that customer documents remain within the European Union. No customer document content is intentionally transferred to external generative AI providers.

11.2 Transfer Mechanisms

Where a service provider is part of a global corporate group, Veltrixa performs vendor due diligence and documents applicable transfer mechanisms, including Data Processing Agreements and, where required, Standard Contractual Clauses (SCCs).

11.3 Safeguards

Safeguard	Purpose	Status
Data Processing Agreement	Processor obligations	Implemented
Vendor Review	Annual assurance	Planned annually
Transfer Impact Assessment	Assess international transfers	Performed where applicable
Encryption	Protect confidentiality	Implemented

12. Privacy Assessment

12.1 Privacy Observations

The reviewed architecture demonstrates several positive privacy characteristics:

- AI processing remains within the EU.
- Documents and metadata are separated.

- Encryption is applied both in transit and at rest.
- Human review prevents fully automated business decisions.
- Temporary processing copies are automatically deleted.

12.2 Recommended Controls

- Maintain Privacy by Design reviews for significant architectural changes.
- Review vendor assurance documentation annually.
- Validate retention configuration during periodic governance reviews.
- Review privileged access on a scheduled basis.
- Continue evaluating opportunities for data minimisation before AI processing.

12.3 Open Risks

Risk	Likelihood	Impact	Recommended Treatment
Unexpected special category data	Medium	High	User guidance and periodic review
Retention configuration errors	Medium	Medium	Configuration validation
Privileged access misuse	Low	High	Least privilege and audit logging
Regulatory changes	Medium	Medium	Annual legal review

12.4 Overall Assessment

Overall, the AI Document Classification capability demonstrates a mature privacy posture. The solution incorporates Privacy by Design principles through secure processing, role-based access control, EU-based hosting, separation of document content and metadata, and controlled retention practices. Continued governance should focus on vendor assurance, international transfer assessments and periodic reviews of access rights and retention settings.

Appendix A – Data Flow Diagram

The complete Personal Data Flow Diagram is presented in **Figure 2** within Chapter 4. It illustrates the end-to-end lifecycle of personal data as it moves through the Veltrixa Cloud AI Document Classification capability, from document upload to secure deletion.

The diagram serves as a visual reference for the processing activities documented throughout this report and supports the detailed analysis provided in Chapter 4.

Appendix B – Supporting Data Inventory

The detailed inventory maintained in Chapter 5 serves as the authoritative inventory for this processing activity.

Appendix C – Glossary

Term	Definition
DFM	Data Flow Mapping
DFD	Data Flow Diagram
DPIA	Data Protection Impact Assessment
ROPA	Record of Processing Activities
RBAC	Role-Based Access Control
SCC	Standard Contractual Clauses
TIA	Transfer Impact Assessment
PbD	Privacy by Design